Higher Diploma in Science in Computer Science

2025-2027

Rosanne Birney
Database

David Drohan
Mobile App

Jimmy McGibney
Devops
Security

Siobhan Roche
Programming

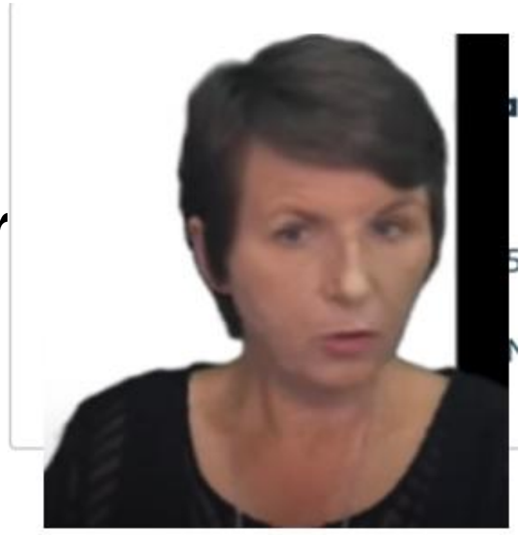
Caroline Cahill
Computer Systems

Richard Frisby
Devops

Laura McGibney
Student
Engagement

Frank Walsh
Computer System
Full Stack 2

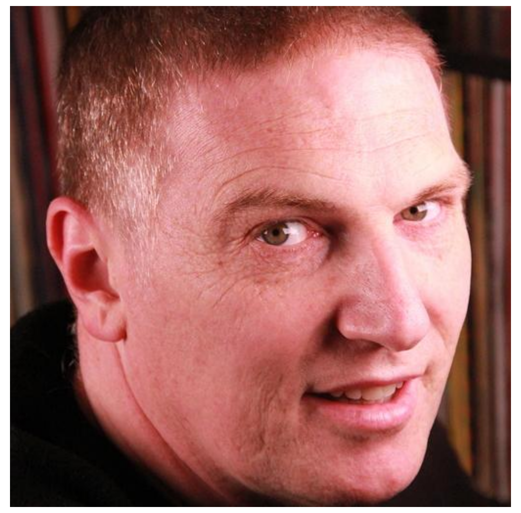
Eamonn de Leastar
Full Stack 2



Mary Lyng
Database

Mairead Meagher
Programming

Peter Windle
Programming



Colm Dunphy
Project

Joan Mangan
Programme &
Placement
Coordinator

John Rellis
Web Dev 1
Web Dev 2

Mujahid Tabassum
Computer System

Agenda

1. Context & Objectives
2. Programme Structure
3. Semesters & Modules
4. Calendar, Timetable & Assessment Sequencing
5. Module Summaries

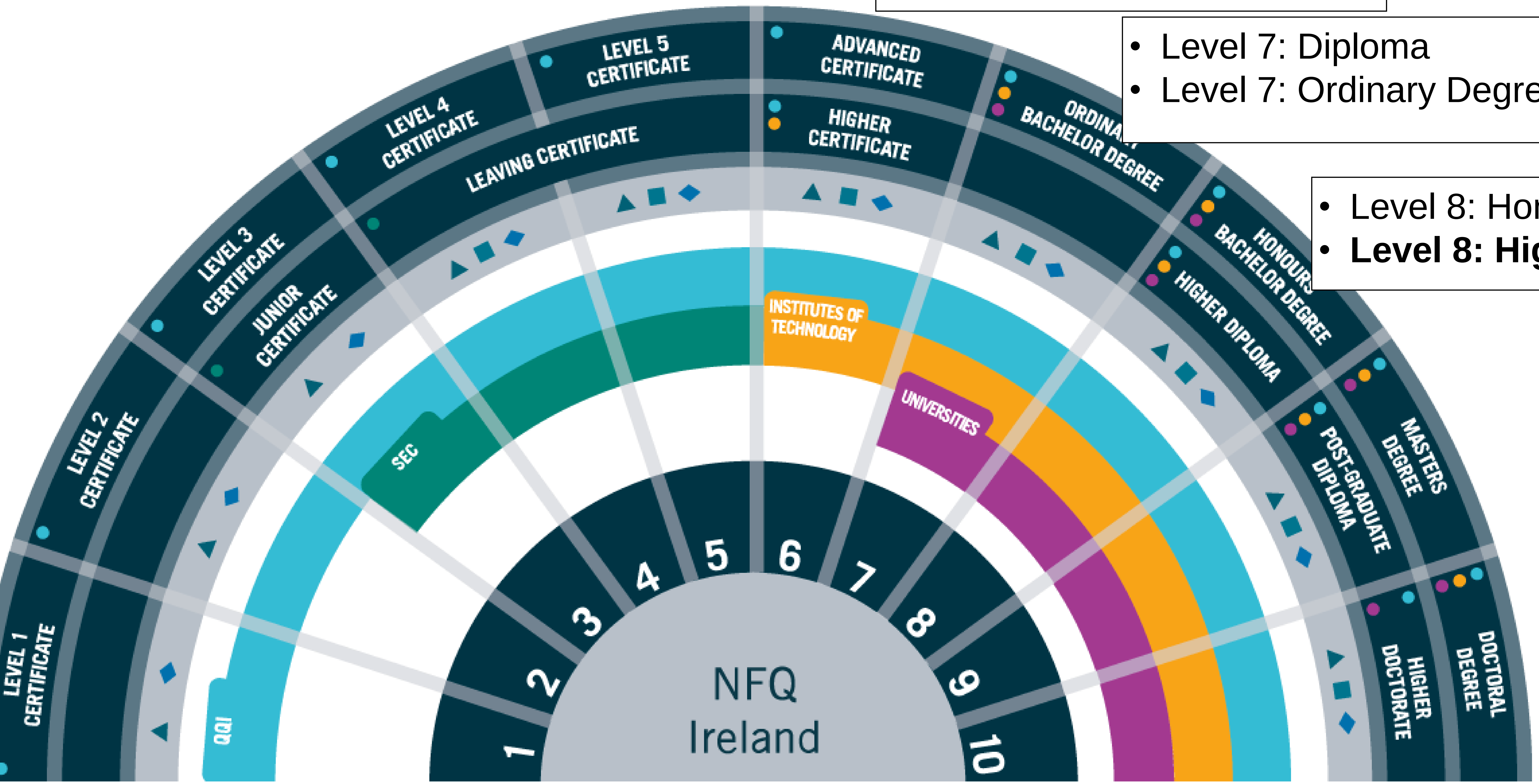
1. Context & Objectives

National Framework of Qualifications / Levels

- Level 6: Certificate
- Level 6: Higher Certificate

- Level 7: Diploma
- Level 7: Ordinary Degree

- Level 8: Honours Degree
- **Level 8: Higher Diploma**



https://nfq.qqi.ie/

ADVANCED CERTIFICATE	ORDINARY BACHELOR DEGREE	HONOURS BACHELOR DEGREE	MASTERS DEGREE	DOCTORAL DEGREE
HIGHER CERTIFICATE		HIGHER DIPLOMA	POST-GRADUATE DIPLOMA	HIGHER DOCTORATE

Key Programme **Features**

- Immersion
- Specialisation
- Industry Partnership

Immersion in Computing Knowledge

*“The participants will be **graduates** who have already obtained significant **transferable skills** by comparison with other undergraduate students...”*

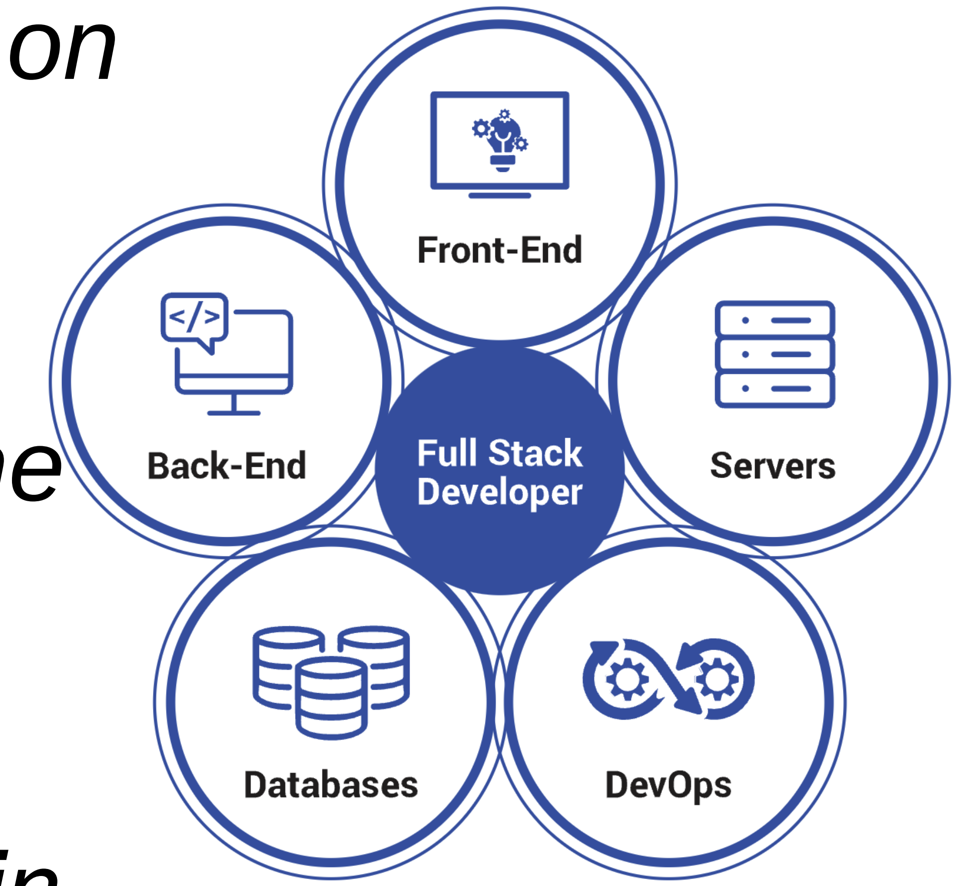
*“**Semester 1** participants will undertake a broad immersive set of modules in the **fundamentals** of computing...”*

*“The **pace** of delivery will have to be **significantly higher** than for normal undergraduate programmes...”*

**‘ICT and Software Development Skills Programme’ HEA Call for Proposals / Terms & Conditions, November 2012*

Deepening and Specialisation

*"In **semester 2 ... a specialisation** which reflects their own strengths as demonstrated on the programme to date..."*



*".. a focused set of modules and project-work designed to bring candidates quickly to the industry entry standard ..." **Junior Software Developer (Full Stack Oriented)***

"Participants will be expected to select their specialisation based on their achievement in semester 1 and their own ambitions..."

**'ICT and Software Development Skills Programme' HEA Call for Proposals / Terms & Conditions, November 2012*

Industry experience and professional development

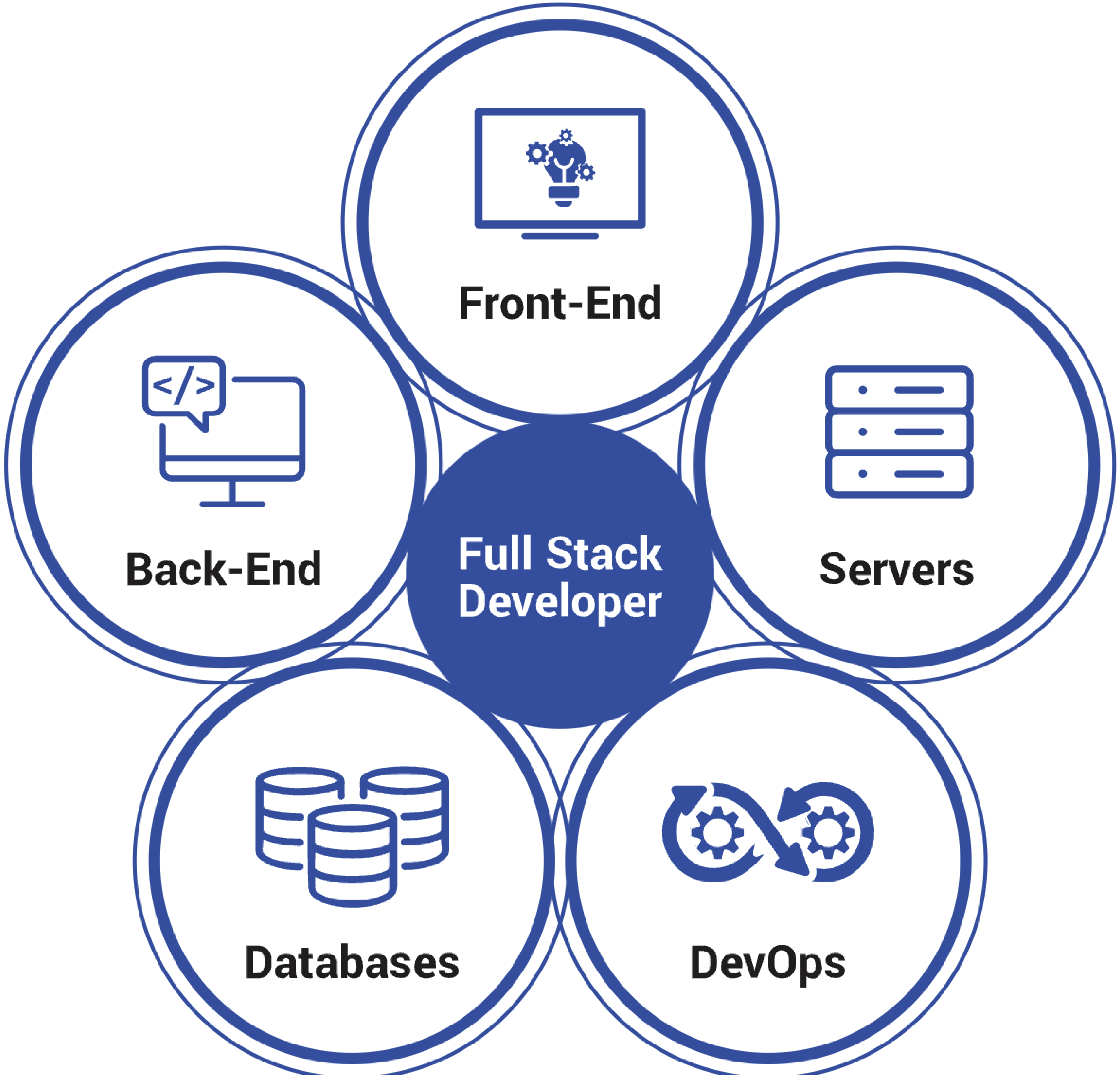
*"**Internships** or **work placements** are seen as crucial to providing graduates with the context and confidence in their new knowledge..."*

"Outputs expected from the work placement would include a work placement report, a project ideally conducted in the work placement organisation..."

*"...academic and **industry partners** will cooperate in the provision of appropriate academic supervision resources for the duration of this work placement activity..."*

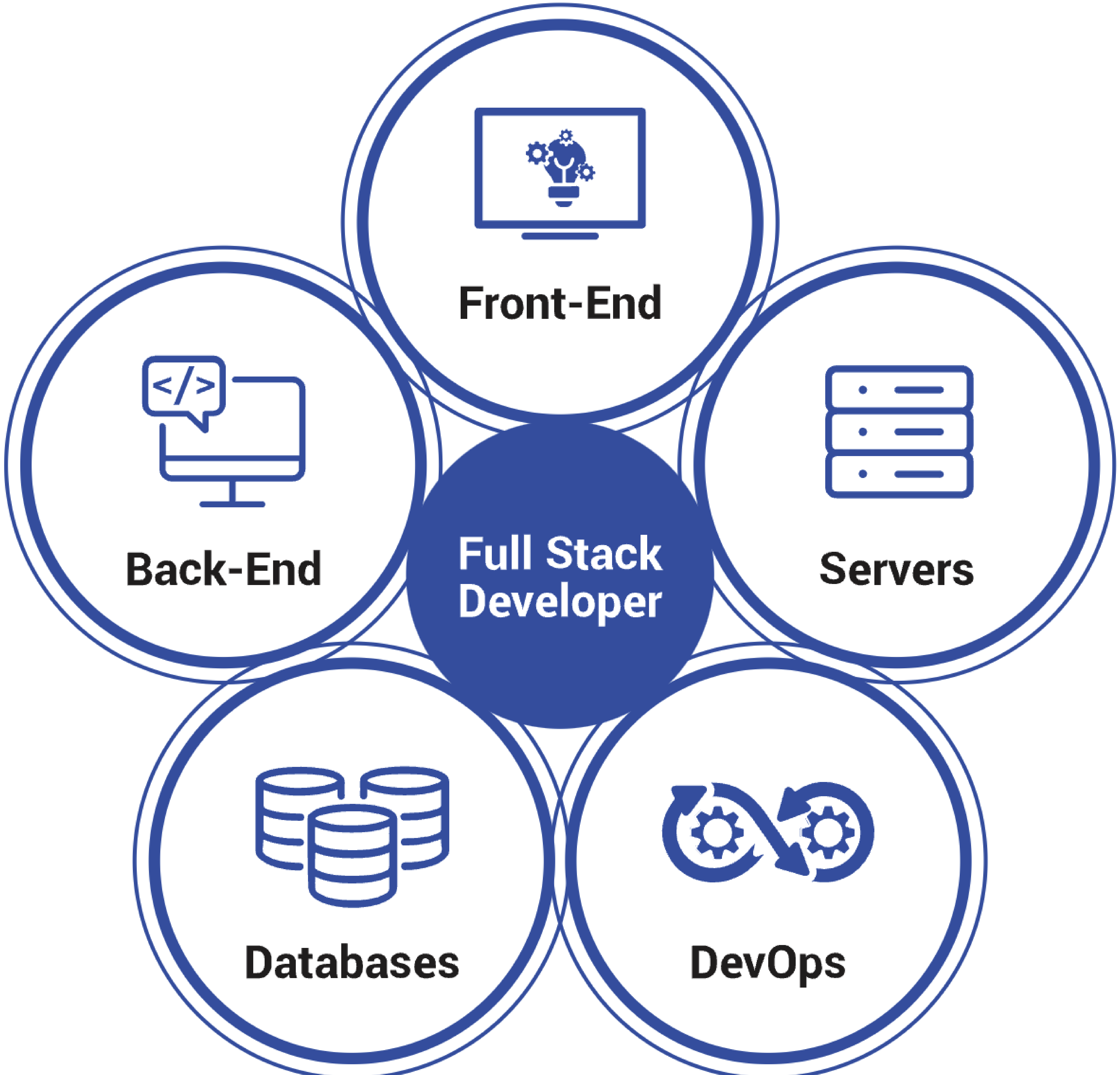
**'ICT and Software Development Skills Programme' HEA Call for Proposals / Terms & Conditions, November 2012*

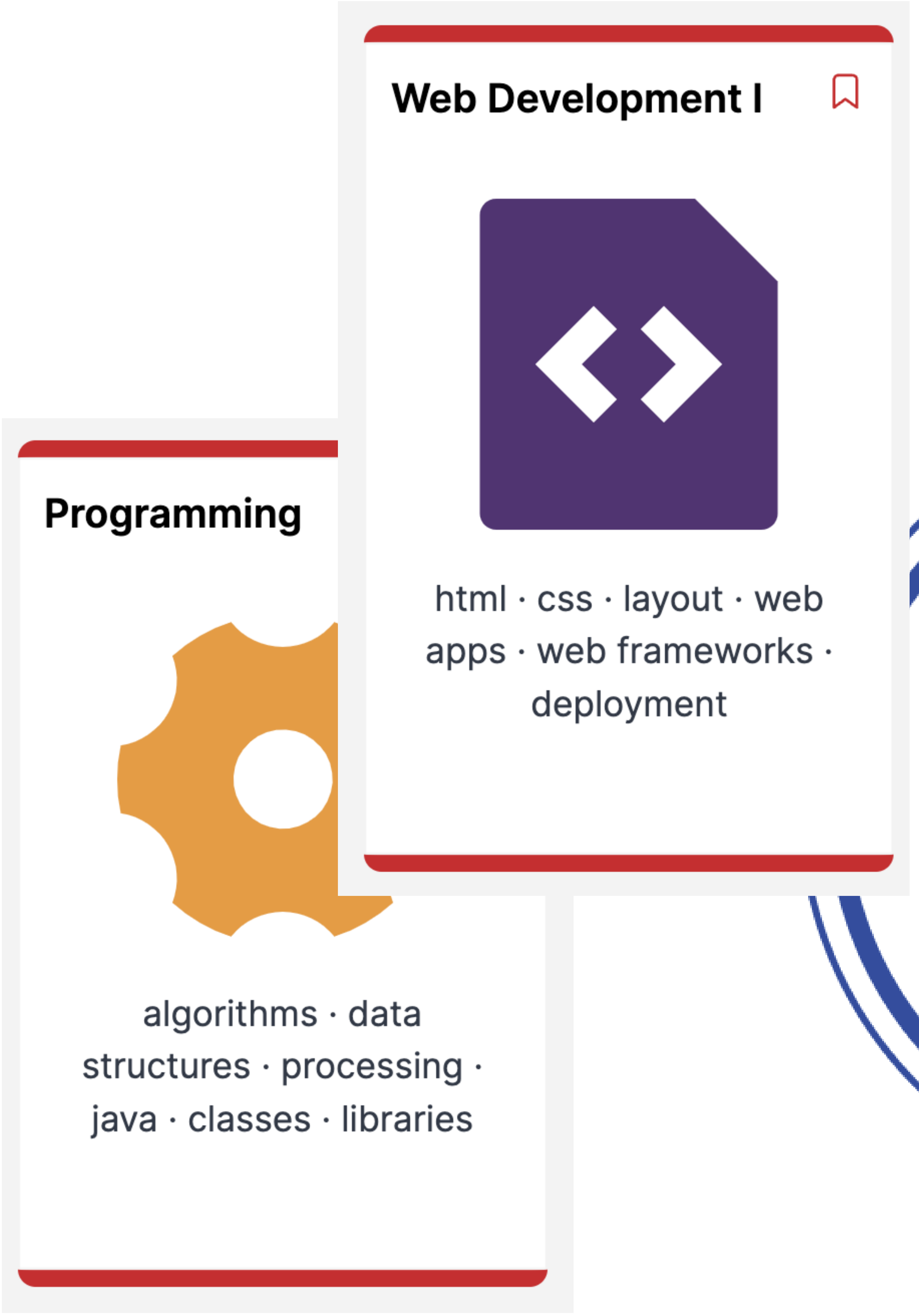
2. Programme Structure

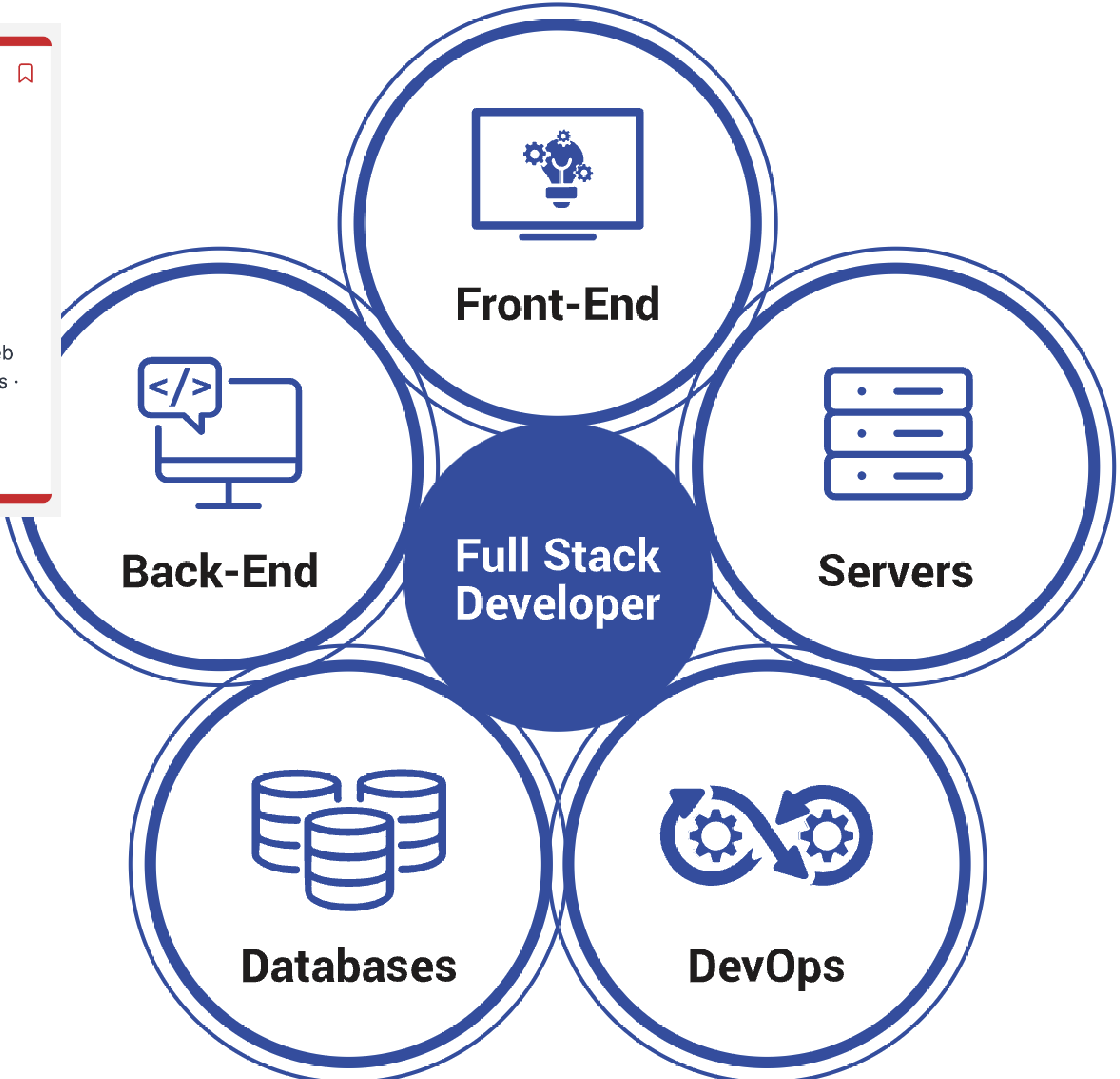

Front-End
Back-End
Full Stack Developer
Servers
Databases
DevOps

Programming

algorithms · data structures · processing · java · classes · libraries




Web Development I
html · css · layout · web apps · web frameworks · deployment
Programming
algorithms · data structures · processing · java · classes · libraries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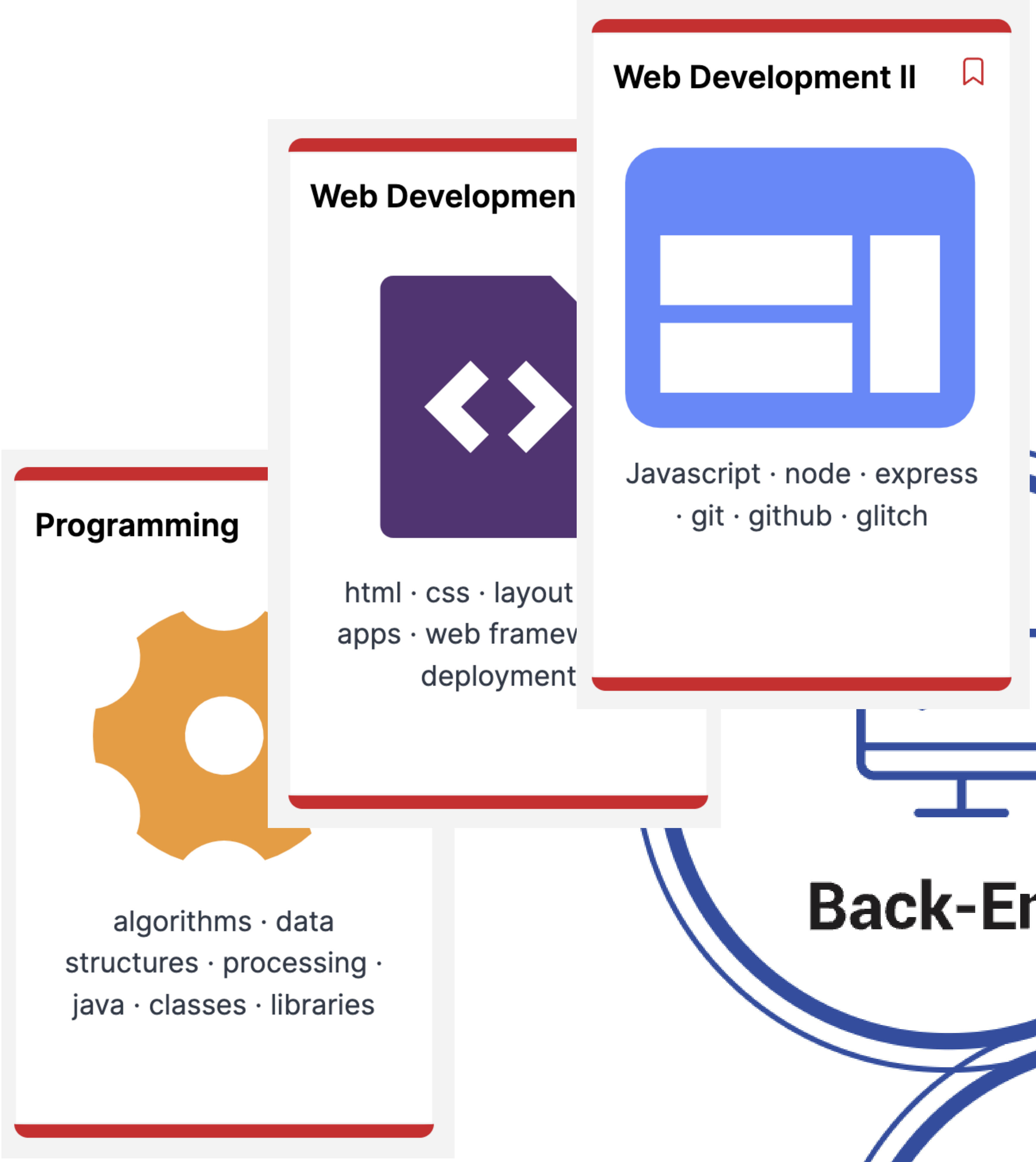
Programming
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Web Developmen
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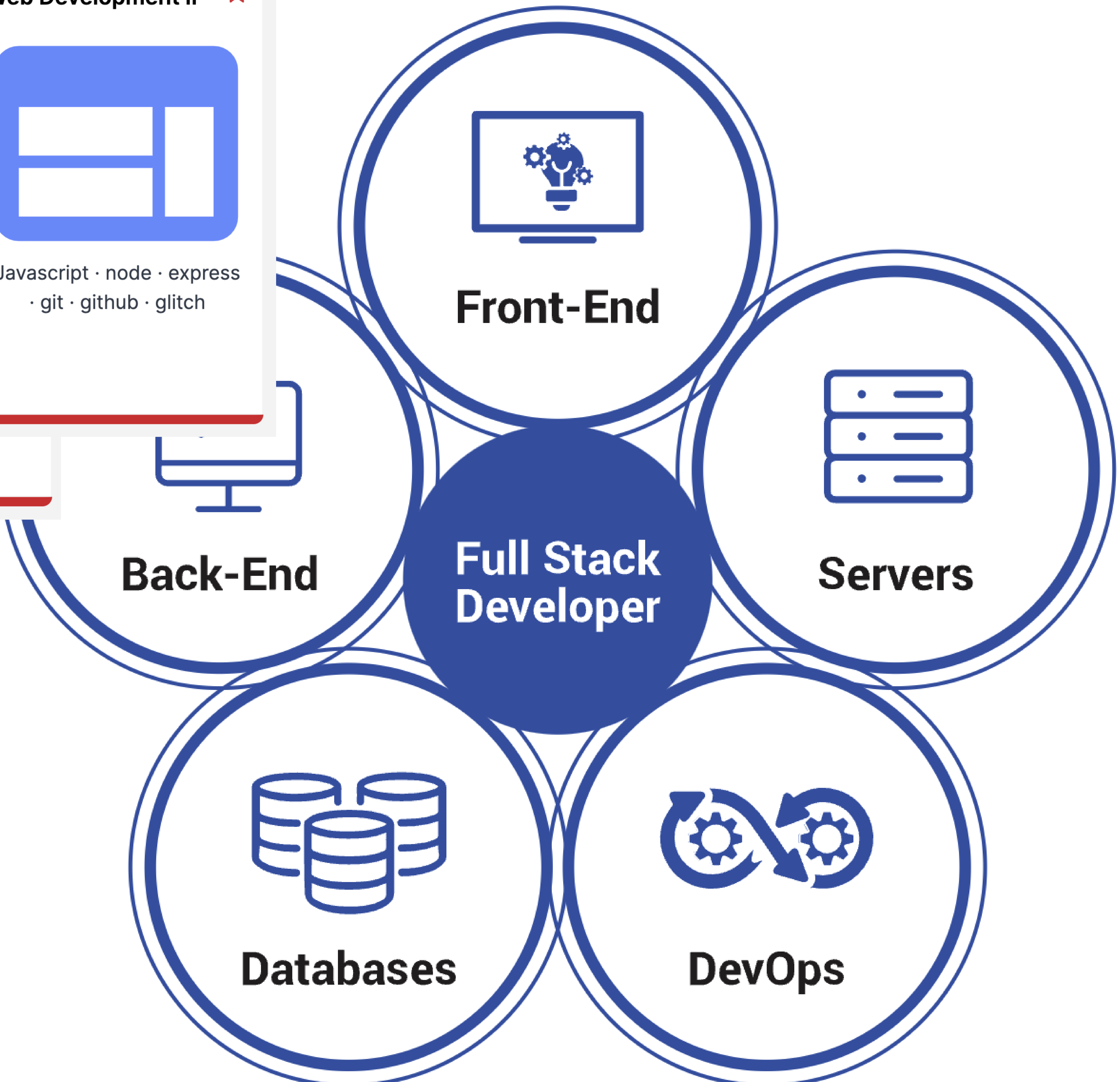
Web Development II
Javascript · node · express · git · github · glitch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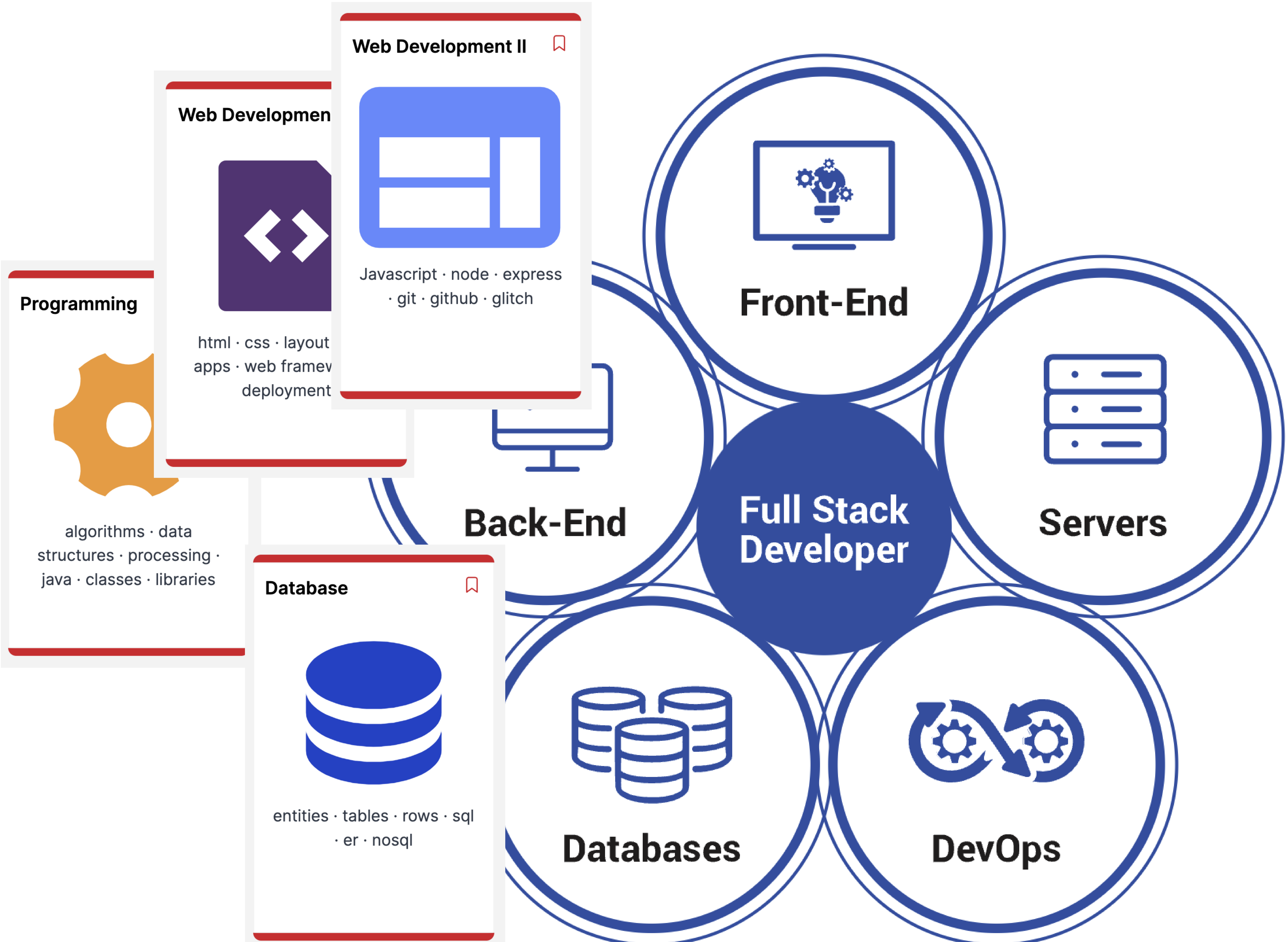
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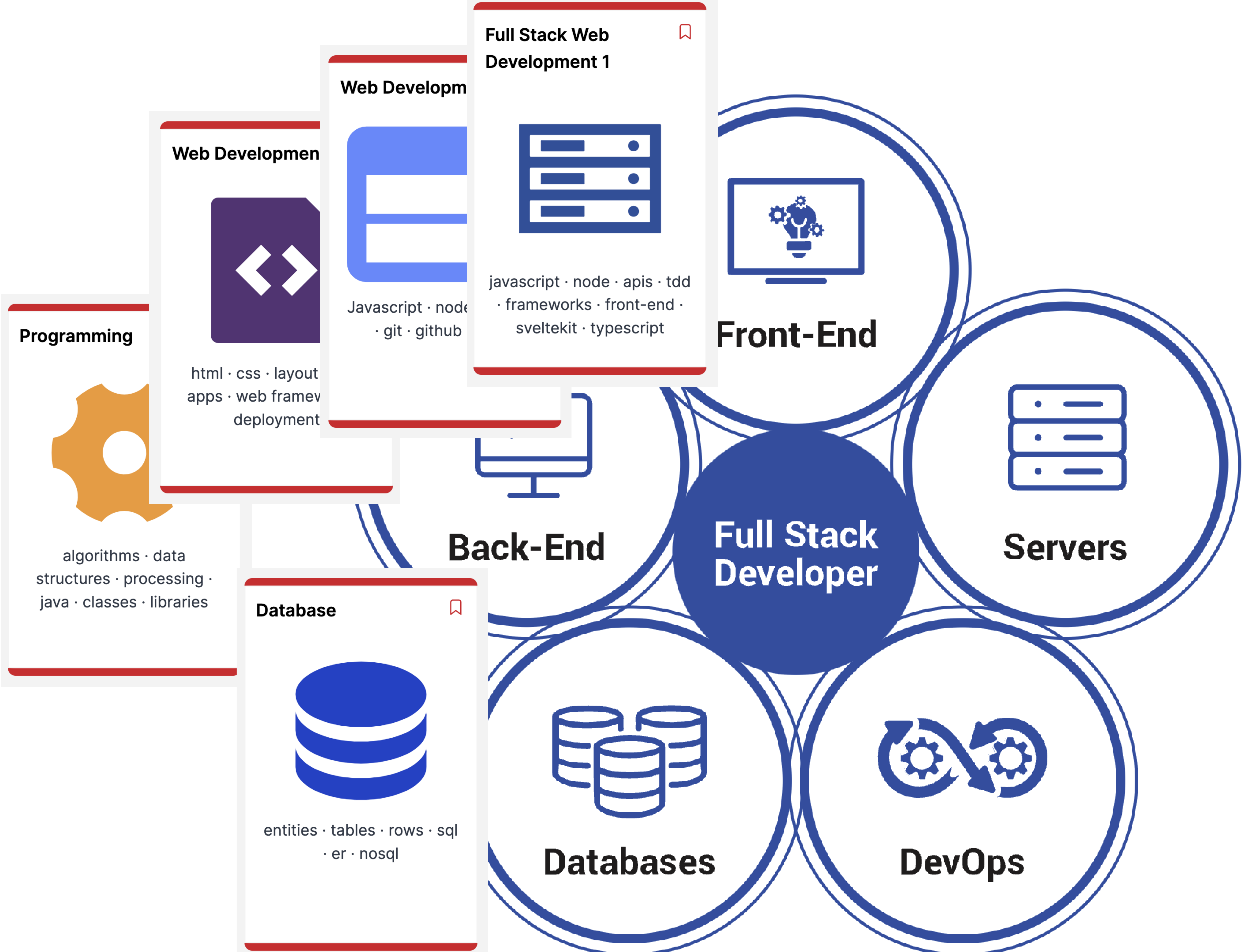
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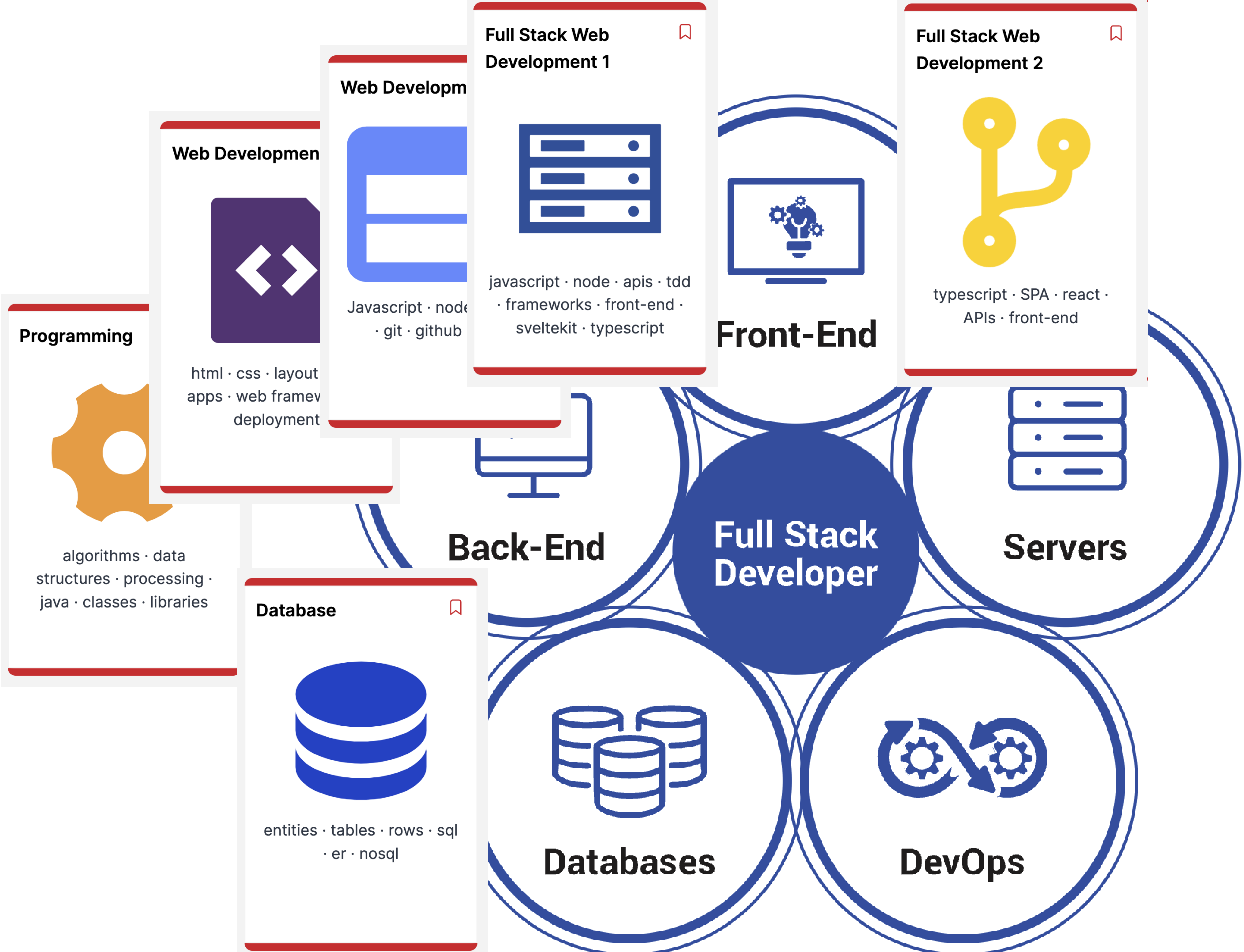
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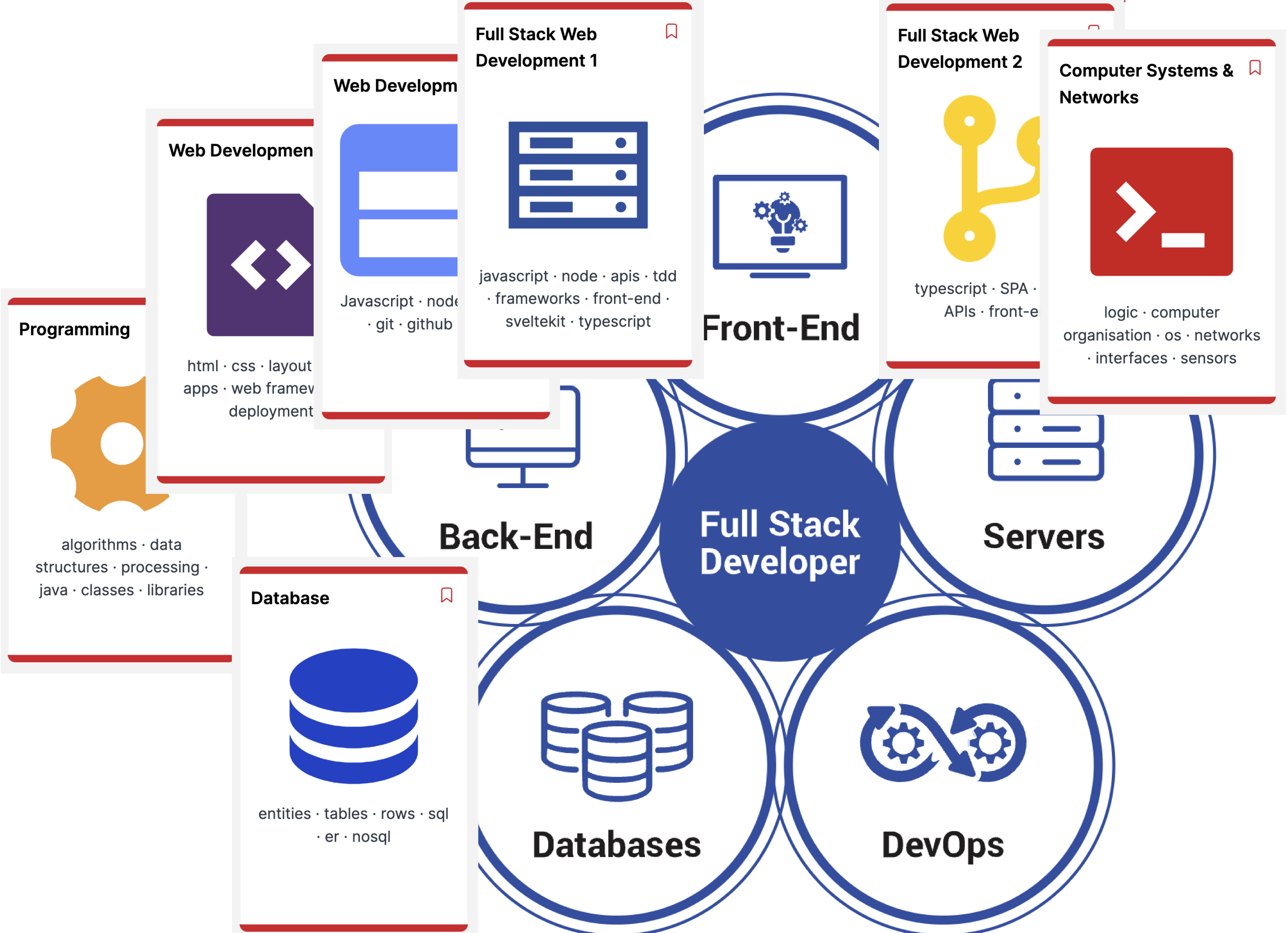

Full Stack Web Development 1
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Full Stack Web Development 2
typescript · SPA ·
APIs · front-e
Computer Systems & Networks
logic · computer organisation · os · networks · interfaces · sensors
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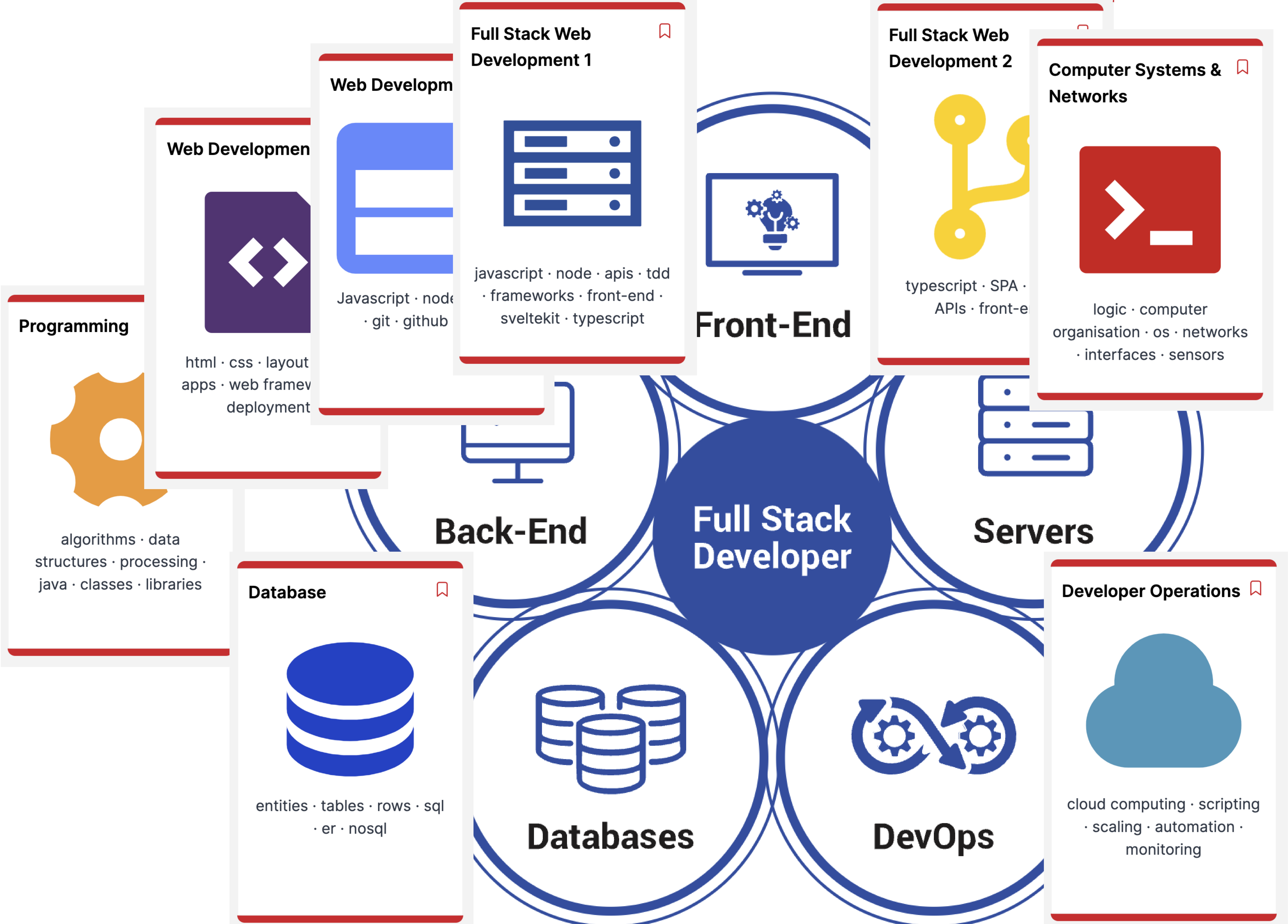
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Mobile App Development
layouts · activities · resources · lifecycle · widgits · ux
Back-End
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cloud computing · scripting · scaling · automation · monitoring
Software Security & Forensics
security · crypto · threats · vulnerabilities · file systems · Operating Systems

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Back-End
Project & Placement
CV Preparation · placement · project handbook · showcases · samples
Servers
Database
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Full Stack Web Development 1
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Full Stack Web Development 2
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Computer Systems & Networks
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Mobile App Development
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Web Develop
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html · css · layout
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Front-End
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Project & Placement
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Back-End
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Developer Operations
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Software Security & Forensics
10
Databases
DevOps
Module Credits


Front-End
-End
Servers
Databases
DevOps
Workshop One
induction · structure · schedules · handbook
Workshop Two
Introducing semester 2
Life etc...
time · space · balance · experiences · tips · techniques
Workshop Three
agile methods · cv preparation · introducing semester 3
Workshop Four
Introducing semester 4, Project, Placement
Workshop Five
Friday January 12th, 2024

3. Semesters & Modules

Semester 1:
Jan-June 2025

1

Semester 2:
June-Dec 2025

2

Semester 3:
Jan-June 2026

3

Semester 4:
Sept-Dec 2026

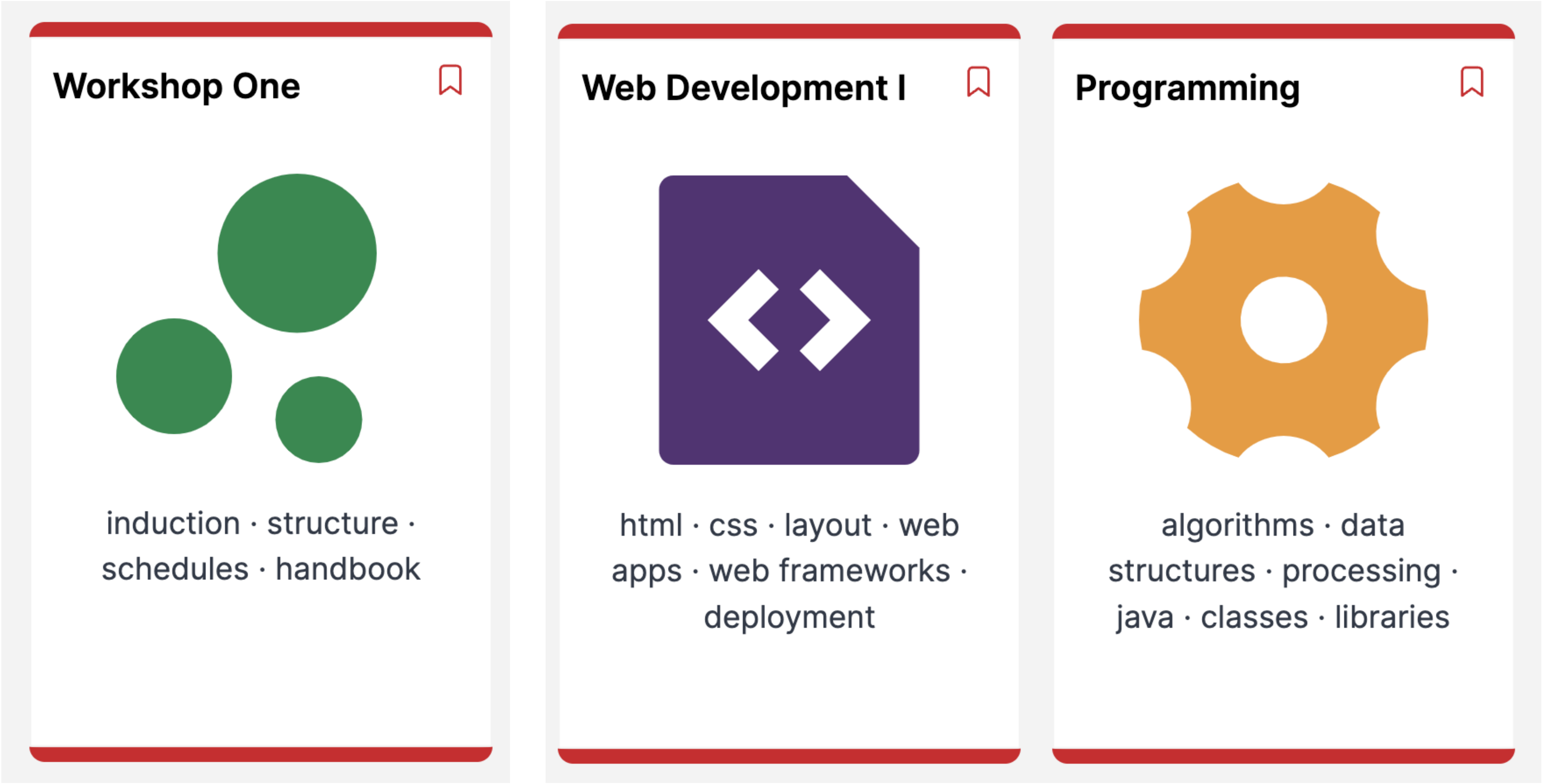
4

June 2026 - May 2027

Project/Work Placement

(Work Placement 4-6 months)

Semester 1: January - June 2025



“..a broad immersive set of modules in the fundamentals of computing covering ***software development, systems analysis & testing****, databases, architecture, OS & networking,* ***web design / user-experience..****”*

Semester 1: June - December 2025

Workshop Two

Introducing semester 2

Web Development II

Javascript · node · express · git · github · glitch

Computer Systems & Networks

logic · computer organisation · os · networks · interfaces · sensors

Database

entities · tables · rows · sql · er · nosql

"..a broad immersive set of modules in the ***fundamentals of computing*** *covering software development, systems analysis & testing,* ***databases, architecture, OS & networking****, web design / user-experience.."*

Semester 3: January - June 2026

Workshop Three

agile methods · cv preparation · introducing semester 3

Full Stack Web Development 1

javascript · node · apis · tdd · frameworks · front-end · sveltekit · typescript

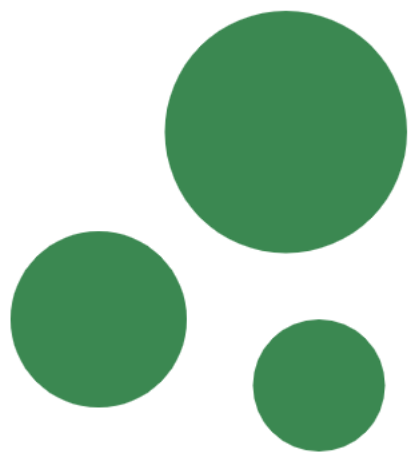
Developer Operations

cloud computing · scripting · scaling · automation · monitoring

"… students are expected to take a specialisation which reflects their own strengths as demonstrated on the programme to date..."

Semester 4: June - December 2026

Workshop Four



Introducing semester 4, Project, Placement

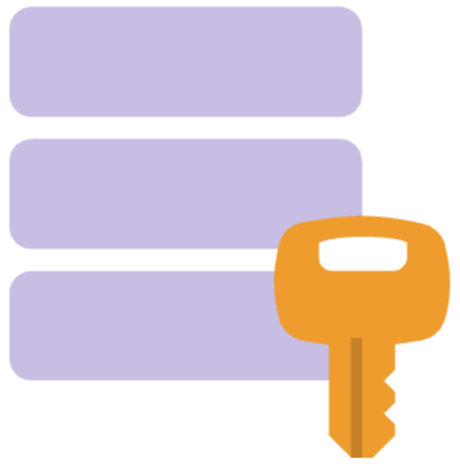
Full Stack Web Development 2

typescript · SPA · react · APIs · front-end

Mobile App Development

layouts · activities · resources · lifecycle · widgits · ux

Software Security & Forensics



security · crypto · threats · vulnerabilities · file systems · Operating Systems

"… students are expected to take a specialisation which reflects their own strengths as demonstrated on the programme to date…"

Semester 4: September 2026 - May 2027

Workshop Five

Friday January 12th, 2024

Project & Placement

CV Preparation · placement · project handbook · showcases · samples

“Internships or work placements are seen as crucial to providing graduates with the context and confidence in the

4. Calendar, Timetable & Assessment Sequencing

HDIP 25

Semester 1 2025

Programming

algorithms · data structures · processing · java · classes · libraries

Web Development I

html · css · layout · web apps · web frameworks · deployment

Web Development 2

git · javascript · glitch · api · node · json · express

		S	M	T	W	T	F	S
	Week				1	2	3	4
January		5	6	7	8	9	10	
		12	13	14	15	16	17	
	1	19	20	21	22	23	24	25
	2	26	27	28	29	30	31	1
February	3	2	3	4	5	6	7	8
	4	9	10	11	12	13	14	15
	reading	16	17	18	19	20	21	22
	5	23	24	25	26	27	28	1
March	6	2	3	4	5	6	7	8
	7	9	10	11	12	13	14	15
	reading	16	17	18	19	20	21	22
	8	23	24	25	26	27	28	29
	9	30	31	1	2	3	4	5
April	*10*	6	7	8	9	10	11	12
	Easter1	13	14	15	16	17	18	19
	Easter2	20	21	22	23	24	25	26
	11	27	28	29	30	1	2	3
May	12	4	5	6	7	8	9	10
		11	12	13	14	15	16	17
	13	18	19	20	21	22	23	
	14	25	26	27	28	29	30	31
June	*14*	1	2	3	4	5	6	7
	16	8	9	10	11	12	13	14
	17	15	16	17	18	19	20	21
		22	23	24	25	26	27	28

Workshop One

induction · structure · schedules · handbook

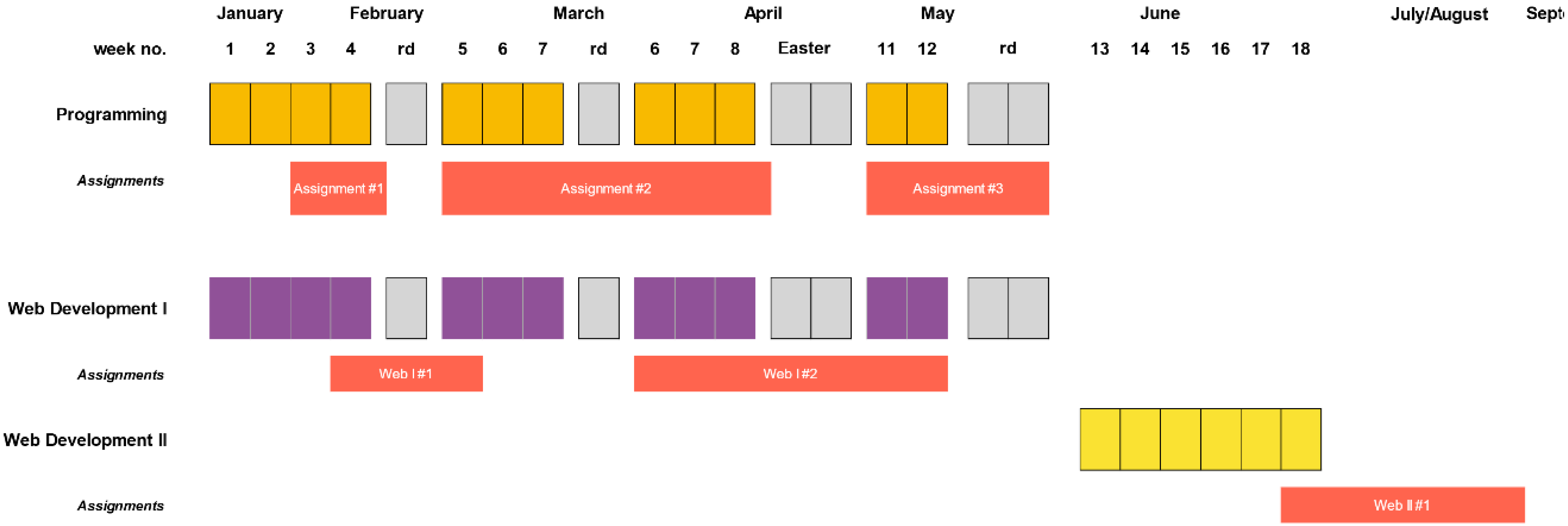
Workshop Two

Introducing semester 2

Weekly Webinar Schedule

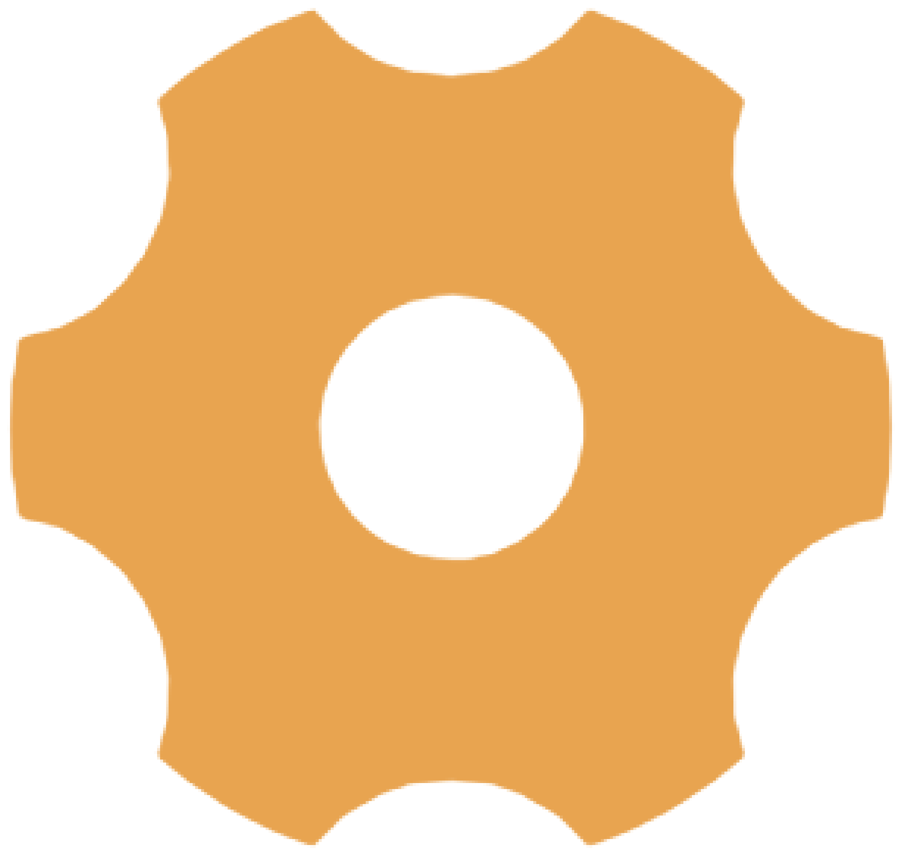
	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	
10:45						10:45
12:15	Programming *Webinar* 12:15-2:00		Programming *Webinar* 12:15-2:00	Web Development *Webinar* 12:15-2:00		12:15
2:00						13:45
15:15						15:15

Semester 1 Assessment Schedule



4. Module Summaries

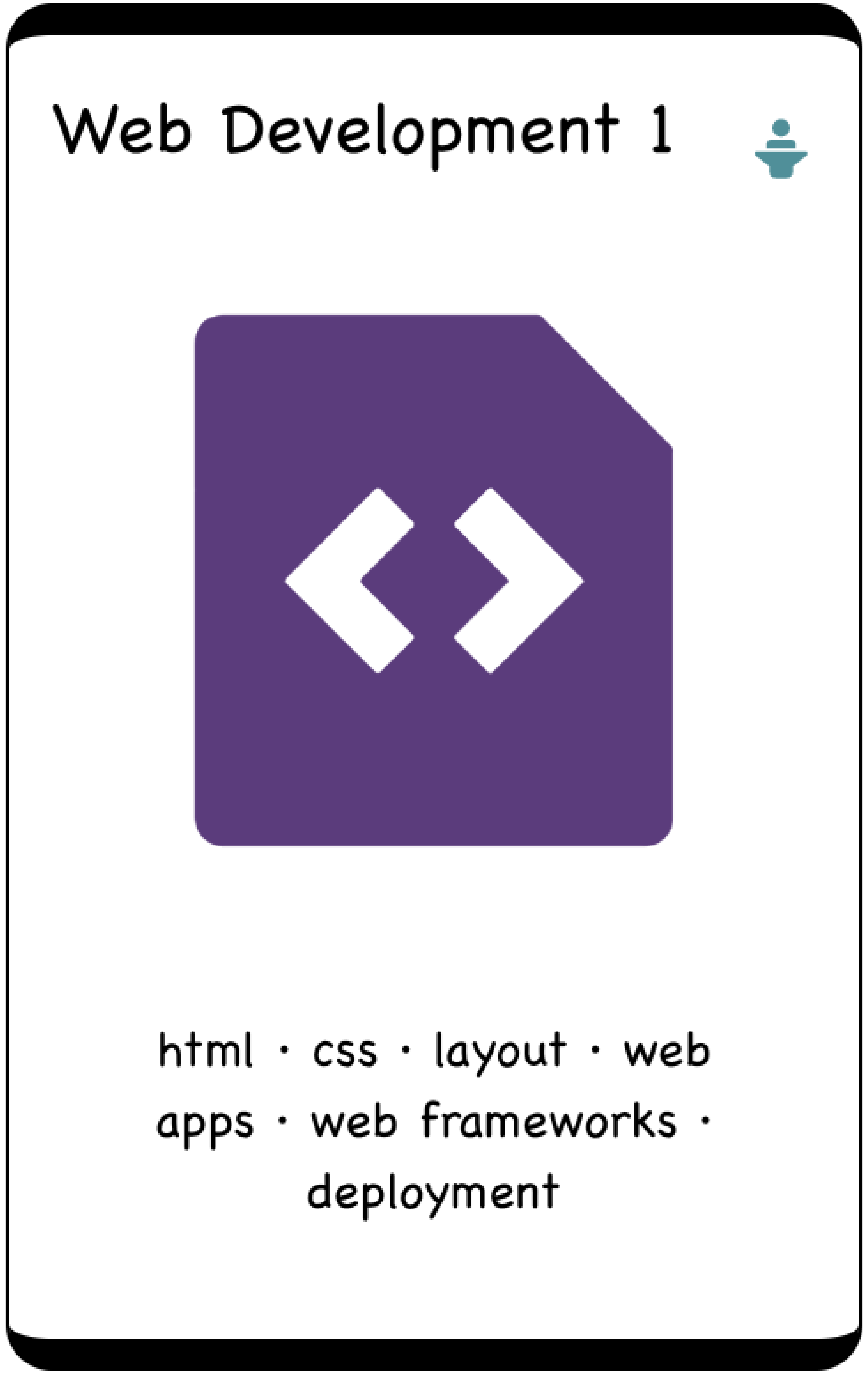
Programming



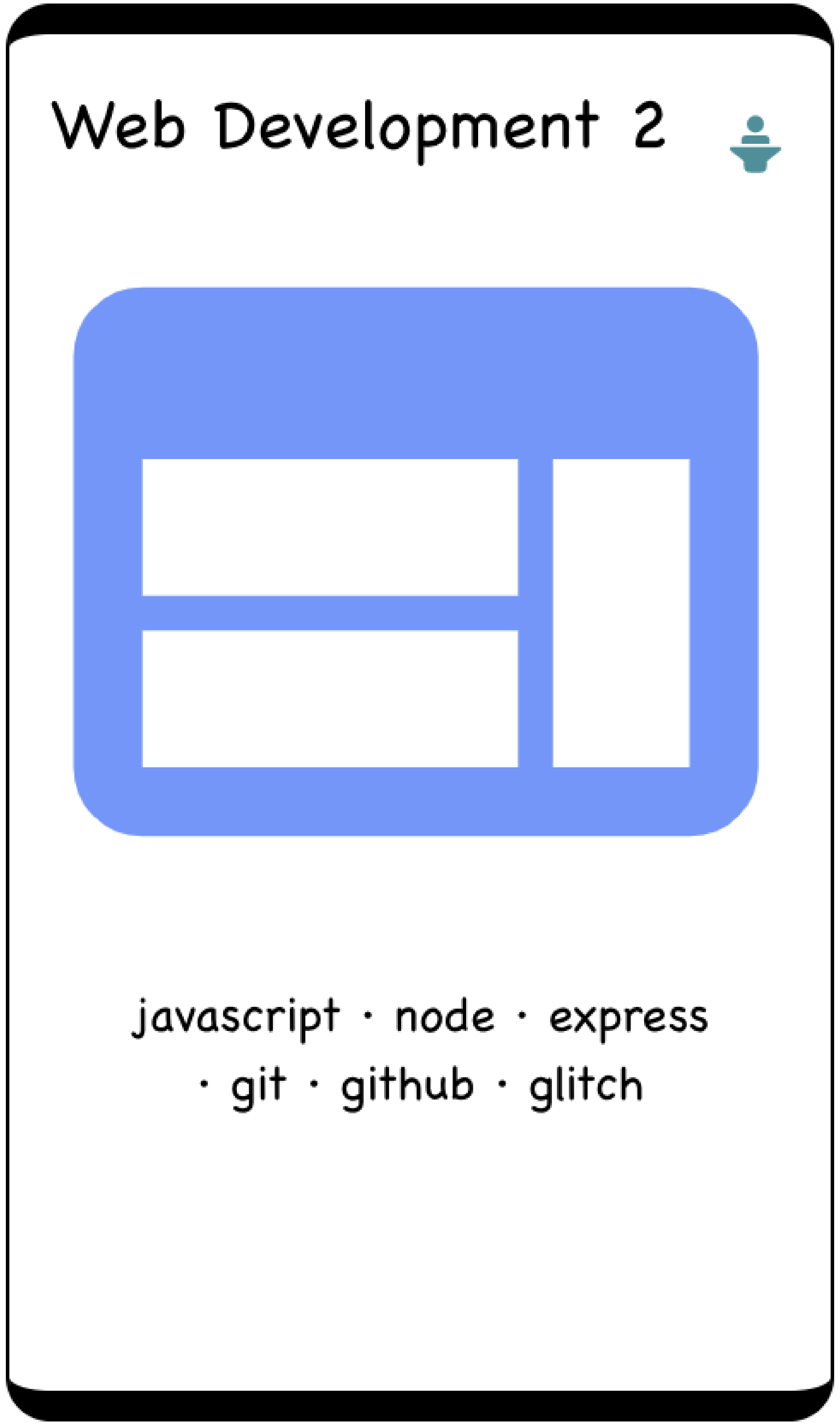
algorithms · data structures · processing · java · classes · libraries

- Apply core problem solving approaches suitable to the programming discipline to build algorithms.
- Construct small applications using standard sequence, conditional and iterative control structures. Change and expand small applications.
- Construct small applications that use simple UI, computation and data structures.
- Apply techniques to effectively test, debug and document small applications.
- Defend and explain how the above applications work.
- Apply problem-solving strategies to various computing problems of increasing complexity.
- Plan, code, test and document applications using advanced programming constructs and data structures

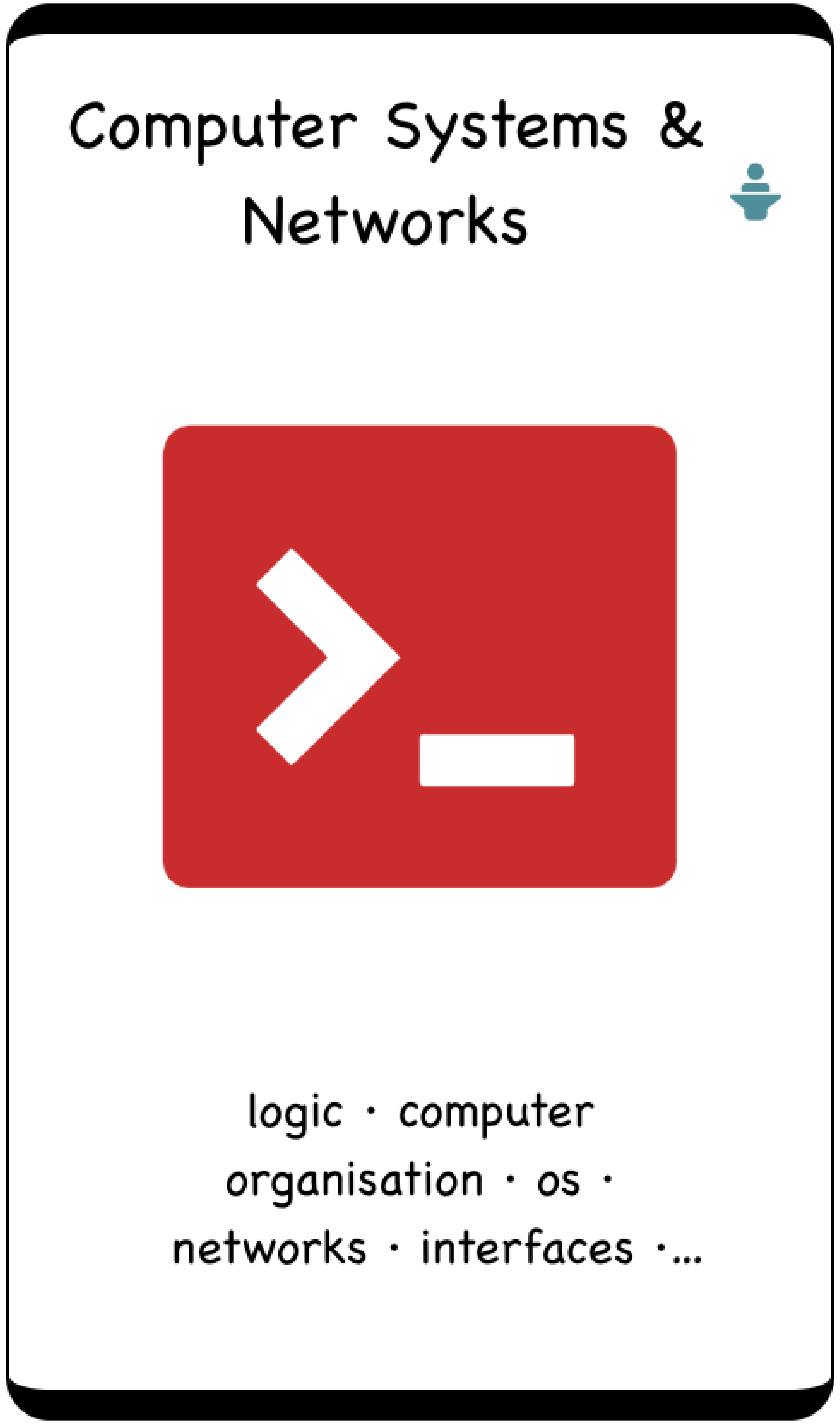
Java



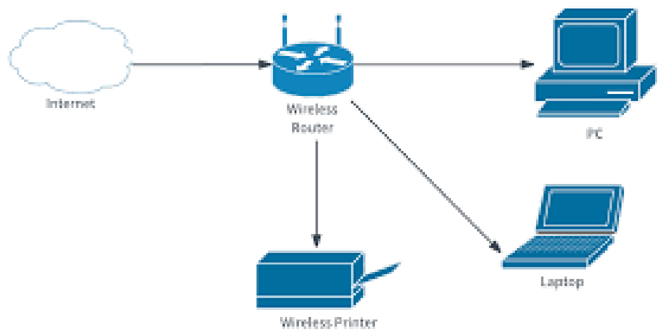
- Understand the fundamentals of the HTML markup language.
- Understand the role of Human Computer Interaction and manipulate CSS to present HTML content.
- Be able to integrate HTML, CSS and Java script to structure simple web sites.
- Understand how a dynamic web page is generated and be familiar with the role of html templating techniques
- Understand the difference between a web site and a web app. Be able to design and implement a simple web app.
- Implement a simple Model View Controller application pattern for a web app.

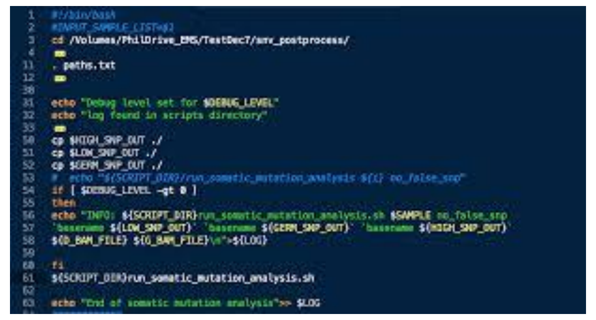


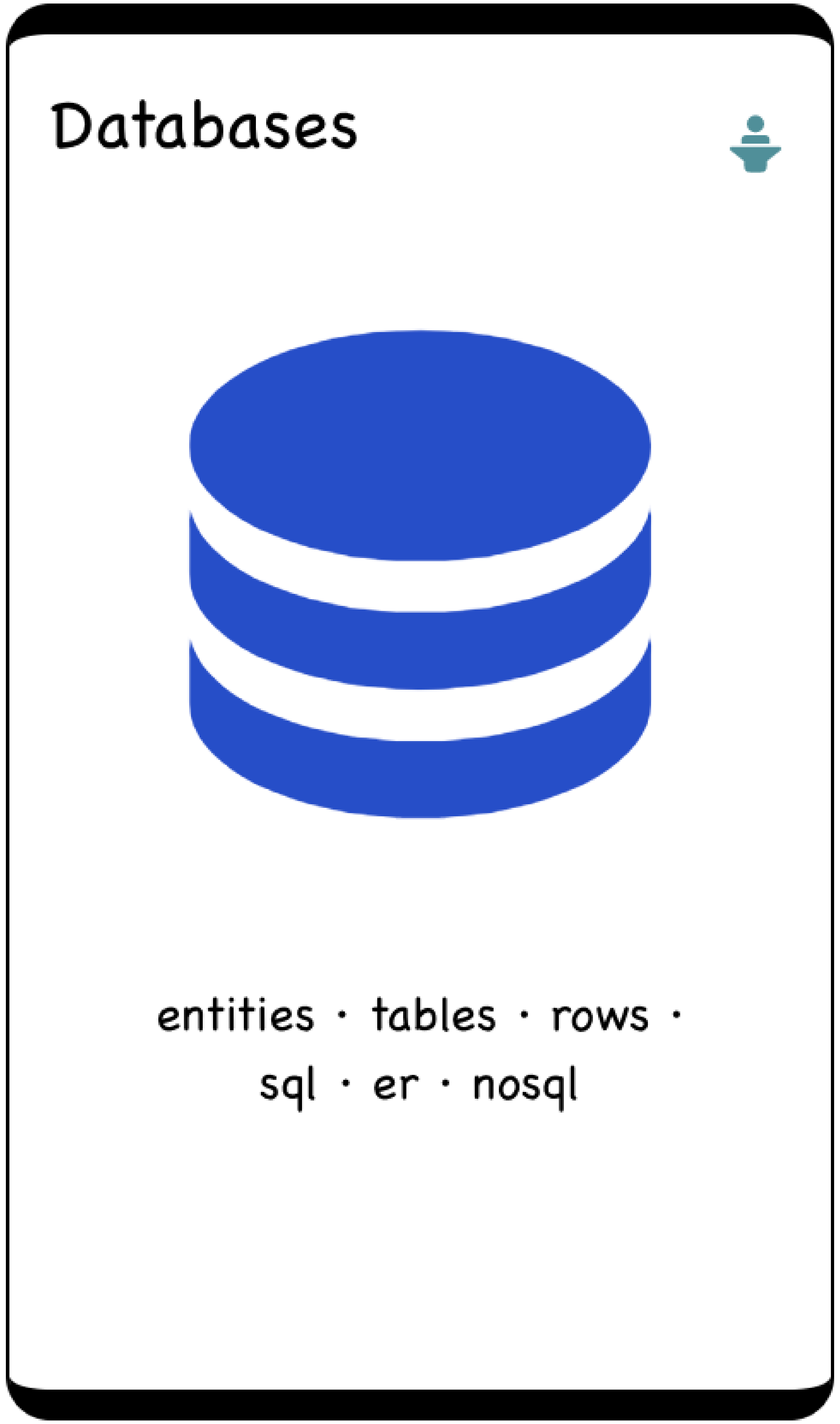
- Continue the journey into web application development
- Establish a competence in Javascript programming language
- Explore the basics of the Node.js framework
- Use a simple JSON persistent storage database
- Design, build and deploy a complete web application using these tools
- Understand the role of Agile methods in this context



- Identify and explain the role various hardware components play in a computer system.
- Use an operating system on a chosen computer architecture.
- Demonstrate an ability to configure systems using the command line.
- Describe the memory management, process management and file management components of a modern operating system.
- Explain basic concepts and theory of networked operating systems and virtualisation.
- Configure a contemporary operating system (within a virtual machine environment)
- Demonstrate competency in a limited set of utilities provided by a contemporary operating system.
- Complete basic automation tasks using scripting.

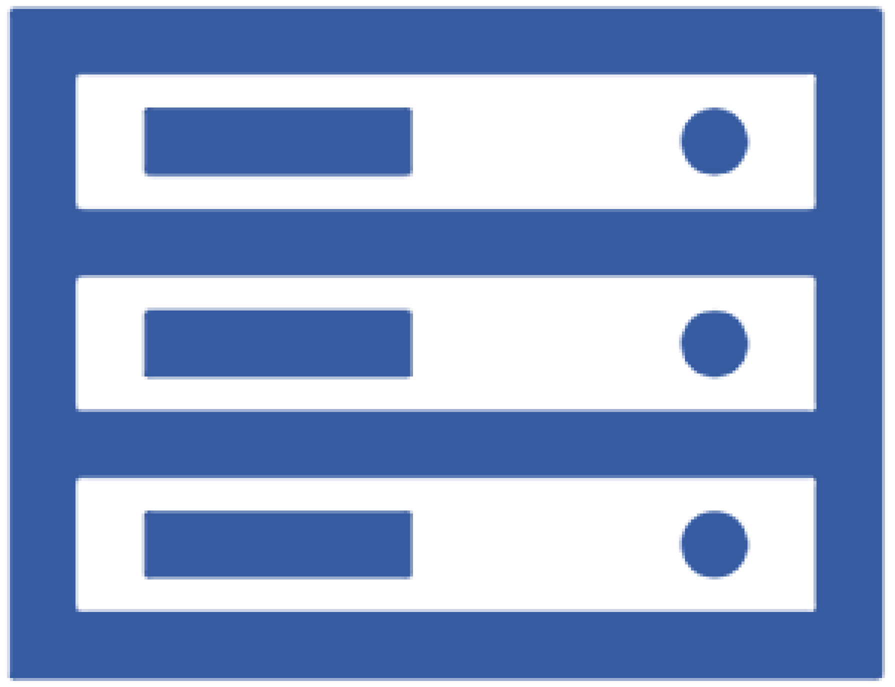





- Discuss the role of a database and its management system.
- Draw Entity Relationship (ER) diagram from an application problem and reproduce this diagram into a set of normalised relations, which are ready for database implementation.
- Design a NoSQL database suitable for a distributed environment with consideration of the CAP theorem.
- Gain an understanding of the physical database design process, its objectives and deliverables.
- Design and implement a database system

Full Stack Development 1



javascript · node · apis · tdd · frameworks · front-end · svelte

- Examine the key components of a server rendered web application and incorporate them into a running application.
- Use Model View Controller & related patterns in the implementation of a web project.
- Relate the request/response lifecycle, routing & session management in the context of a modern application framework.
- Model the user requirements and realize the model in a simple database.
- Apply best practice principles and patterns to the design and documentation of a web API.
- Apply best practice principles and patterns to the design of a medium-sized Single Page Web App.
- Develop an end-to-end web app that supports session management and persistence for a constrained functional requirement set.

Full Stack Development 2

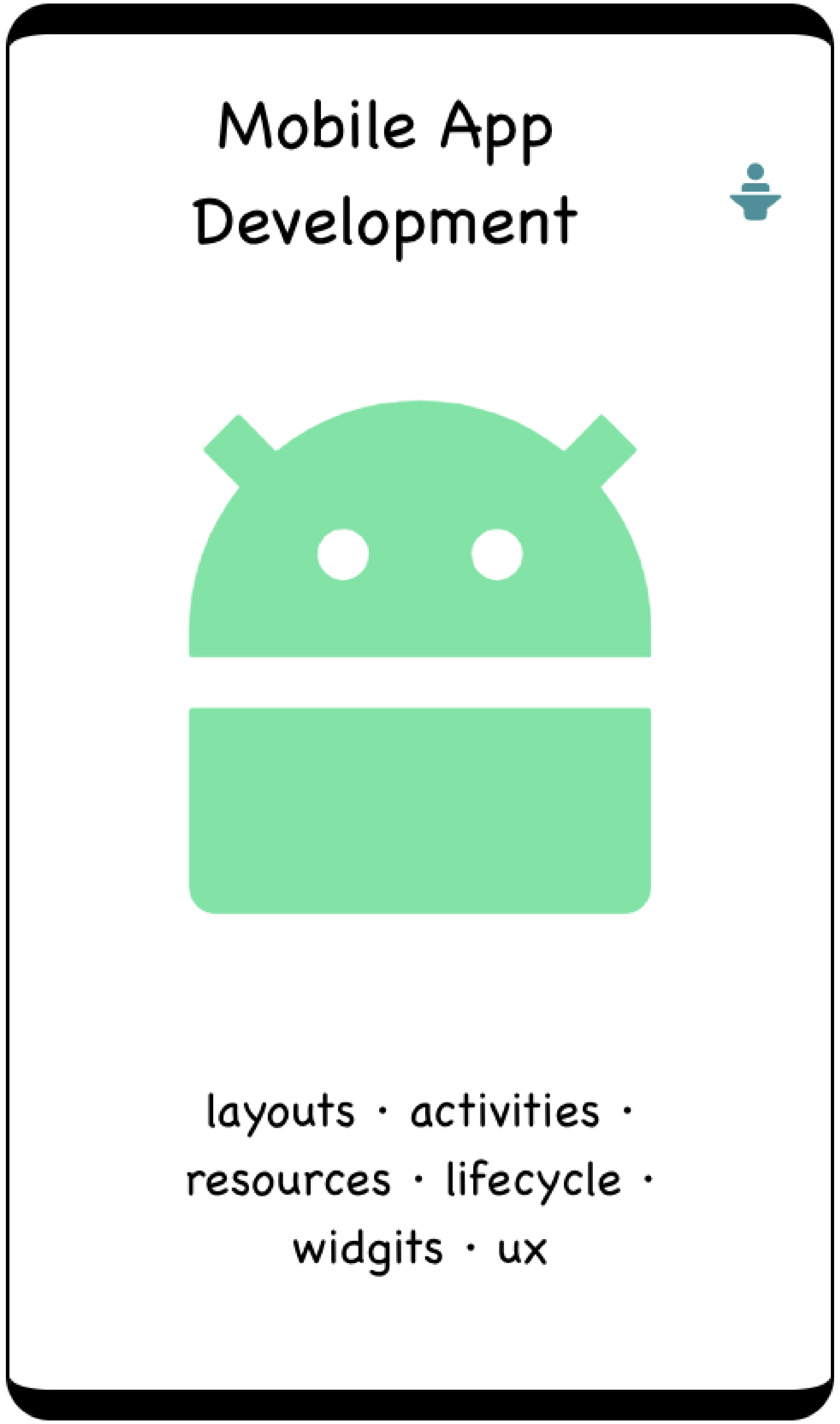
SPA · react · APIs · front-end

- Introduce React + Storybook
- Explore the React component model
- Understand component navigation, lifecycle & routing
- Review the react methodology
- Select appropriate state management strategies & components

Developer Operations

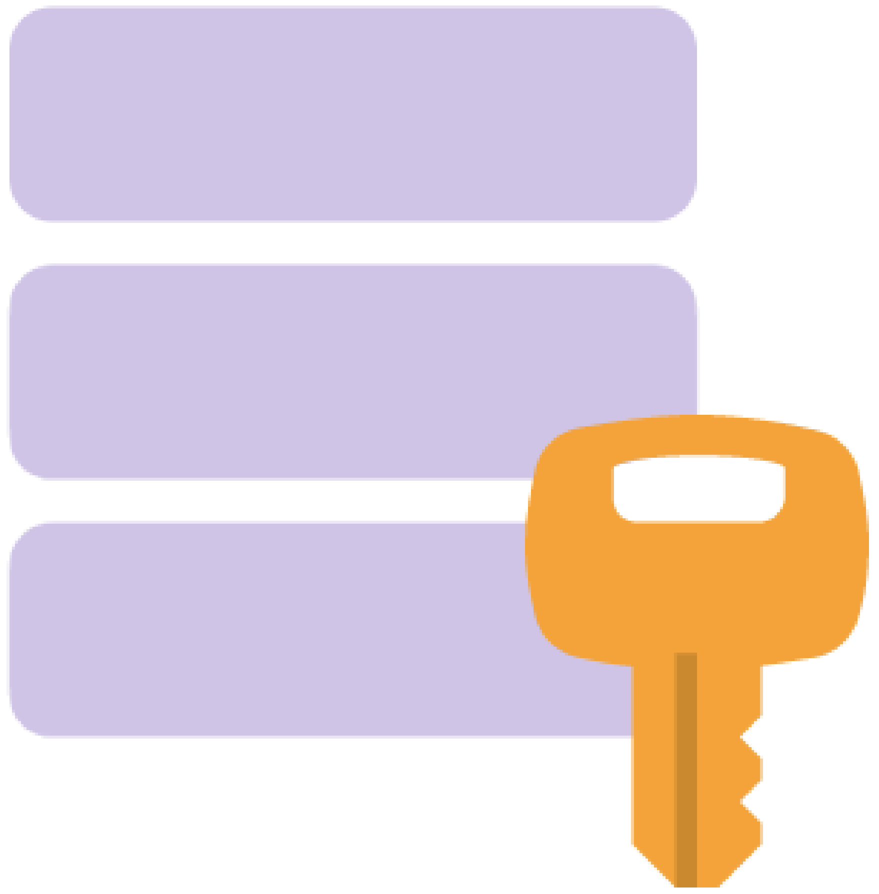
cloud computing · scripting · scaling · automation · monitoring

- Build, configure and manage essential network infrastructure services.
- Build, configure and manage essential application services.
- Deploy a network monitoring solution.
- Develop scripts to assist in the management and automation of modern network services.
- Configure appropriate security mechanisms, including firewall rules, encrypted services, and authentication.



- Decompose an application into its constituent parts, including but not limited to: core application components, user experience resources, packaging.
- Design a coherent User Experience - using appropriate tools, practices and guidelines - for a moderately sized application. Produce a medium sized application, based on a limited set of design patterns.
- Manage the application lifecycle. Structure persistent storage on a device and reliably save and restore application state.
- Select the appropriate design patterns and tools in the development of complex mobile apps.
- Comment on the chosen mobile app framework and the underlying hardware components.
- Design and develop complex multi-screen mobile apps from concept through to completion using best practices and guidelines.
- Set up the interaction of an application with internal sensors and physical subsystems.
- Integrate a remote service API within an application, perhaps based on REST principles, to deliver aspects of its core features set.

Software Security



security · crypto · threats · vulnerabilities

- Demonstrate specific security problems that can arise with web applications and how to address them.
- Compare and contrast alternative approaches to authentication in both enterprise and consumer-oriented
- web applications.
- Use a selection of best security practices in a web application.

GnuPG

OWASP
Open Web Application
Security Project

OpenSSL
Cryptography and SSL/TLS Toolkit

OAUTH

Opportunities for Further Study

- The development team are closely involved in the delivery of two potential follow-on graduate programmes:
  - MSc in Enterprise Software Systems
- These are mature courses, closely aligned with research at the Walton Institute, with substantial enrolments in part-time mode from industry practitioners in the region.
- Successful candidates could continue their academic development in part-time or full-time capacity.

Irish Software Association
ISA Awards 2012
HIGHLY COMMENDED

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Web Developm
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html · css · layout apps · web framew deployment
Programming
algorithms · data structures · processing · java · classes · libraries
Front-End
Back-End
Servers
Databases
DevOps
Full Stack Web Development 2
typescript · SPA · APIs · front-e
Computer Systems & Networks
logic · computer organisation · os · networks · interfaces · sensors
Mobile App Development
layouts · activities · resources · lifecycle · widgits · ux
Project & Placement
CV Preparation · placement · project handbook · showcases · samples
Database
entities · tables · rows · sql · er · nosql
Developer Operations
cloud computing · scripting · scaling · automation · monitoring
Software Security & Forensics
security · crypto · threats · vulnerabilities · file systems · Operating Systems